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Problem introduction

In our project, we focus on predicting the presence of disease based on patients' symptoms. Our goal is to correctly identify patients' disease based on their symptoms. This is a multiclass classification problem - each disease is assigned into one of eight classes. This study was driven by the underlying hypothesis of whether symptoms are sufficient to distinguish between different diseases.

Important findings from the data analysis

While some diseases are easy to identify based on symptoms, some confuse the models. This is because there is not a clear boundary between the diseases - for example, asthma can in some cases present with a severe cough, which is also the main symptom in **cough** and **common cold**.

Model selection

We chose Random Forest as our baseline model because it is robust and can better capture non-linear relationships. Based on the research performed, we decided to select **XGBoost** as the second model, which can be considered as a state of the art model. In existing papers that dealt with similar problems to ours, it performed better than Random Forest.

Results

All developed models achieved over 90% overall accuracy in classifying the eight diseases. While most classes were predicted with high precision and recall, a few diseases - such as asthma, cough and common cold proved more challenging due to overlapping symptoms, which led to occasional misclassifications. To address this, we applied threshold optimization to improve the balance between false positives and false negatives. Each misclassification was analyzed and found to be justifiable based on clinical symptom similarities, ensuring the models remained interpretable and medically reasonable.

Conclusion

In our solution, we concluded that for evaluating overall classification accuracy, models without adjusted decision thresholds are appropriate. However, when optimizing sensitivity and specificity—especially in diagnostically critical contexts—threshold tuning is more suitable. This reflects a trade-off that should be guided by the specific goals of the application.